Ans. to the ques. no. 1

Amdahl's law states that a processor speedup will always be limited by the part of the program that cannot be parallelized. Amdahl's law supports the 'Make the common case faster' because overall system performance is limited by the part that run more frequently if in a program 90% of runtime is for an addition operation and 10% is a subtraction operation. Even if we ~~overcome~~ increase the speed of the subtraction operation, significantly the overall speedup will be smaller. So we should improve the addition operation as it is the most commonly need operation.

Ans. to the ques. no. 2

Throughput is the total work done per unit time, whereas response time is the time between the start and end of the task. Throughput focuses on the system productivity and is important for batch systems. It also measures the quantity of work done per unit time. While on the other hand, response time measures the time taken for one task, focuses on user experience and is important for interactive systems.

One may ~~priotize~~ be prioritized over the other depending on the goals and context. If the main objective is to complete as many tasks as possible within a given time, then throughput should be prioritized. But if the objective is individual user experience and quick feedback, then response time should be prioritized.

Ans. to the ques. no. 3

a

Processor	Clock rate	CPI
P1	4×10^9 Hz	2
P2	2.5×10^9 Hz	1.5
P3	5×10^9 Hz	2.7

We know,

$$\text{Performance} = \frac{1}{\text{Exec Time}}$$

For P1, P_2 & P_3, IC=1 as they follow the same set of instruction.

$$\text{Performance } P_1 = \frac{1}{\frac{CPI}{\text{Clockrate}}} = \frac{1}{\frac{2}{4\times10^9}} = 2\times10^9$$

$$\text{,, } P_2 = \frac{1}{\frac{1.5}{2.5\times10^9}} = 1.67\times10^9$$

$$\text{,, } P_3 = \frac{1}{\frac{2.7}{5\times10^9}} = 1.85\times10^9$$

So P_1 has the highest performance.

(b)

$T_{new} = 0.6\, T_{old}$

$CPI_{new} = 1.35\, CPI_{old}$

Now,

$$\frac{IC \times 1.35 CPI}{T_{new}} = 0.6 \times \frac{IC \times CPI}{T_{old}}$$

or, $$\frac{1.35}{T_{new}} = \frac{0.6}{T_{old}}$$

or, $$T_{new} = \frac{1.35}{0.6} T_{old}$$

$$= 2.25\, T_{old}$$

That is, clock rate must be 2.25 times faster than the origenal clock rate.

Ans. to the ques. no. 4

Computer	Clock Rate	CPI
A	3×10^{9} Hz	1.5
B	2.5×10^{9} Hz	1.2

For computer A & B, IC ~~fotto~~ = 1 as they follow the same set of instruction.

$$\text{CPU Time}_A = \frac{CPI_A}{\text{Clock Rate}_A} = \frac{1.5}{3\times10^{9}} = 5\times10^{-10} = 0.5\,s$$

$$\text{,, ,, }_B = \frac{CPI_B}{\text{Clock Rate}_B} = \frac{1.2}{2.5\times10^{9}} = \cancel{0.48}\ 4.8\times10^{-10} = 0.48\,s$$

That is, Computer A > Computer B.

a

Ans. to the ques. no. 5

50% ALU = 0.5

30% load/store = 0.3

20% branch = 0.2

Total IC = 0.5 + 0.3 + 0.2 = 1

We know,

$$\text{Average CPI} = \frac{\Sigma(\text{CPI of instruction} \times \text{No. of instruction})}{\text{Total IC}}$$

$$= \frac{(0.5 \times 1) + (0.3 \times 2) + (0.2 \times 3)}{1}$$

$$= 1.7$$

Ans. to the ques. no. 6

$IC_{new} = 0.85\, IC$

$CPI_{new} = 0.90\, CPI$

Clock rate remains the same.

We know,

~~CPI = T~~

$$\text{CPU Time / Exec Time} = \frac{IC \times CPI}{\text{Clock Rate}}$$

Since clock rate is same, we don't need to bother about it.

That is,

$$\text{Exec Time}_{new} = 20\,(IC_{new} \times CPI_{new})$$

$$= 20\,(0.85 \times 0.90)$$

$$= 15.3\ s$$